Source-Based AI Recommendation Explanations — Pilot Protocol v1

1. Purpose of the Pilot

The purpose of the pilot study is to evaluate whether the experimental procedure, stimuli, manipulation, and measurement instruments function as intended before launching the main study.

The pilot is not intended to provide definitive evidence for the substantive hypotheses.

The primary objectives are to determine whether:

1. participants understand the shopping scenario and AI recommendation task;
2. the five explanation-source conditions are perceived as intended;
3. the experimental stimuli are sufficiently matched in clarity, tone, and presentation;
4. the survey measures are understandable and show reasonable internal consistency;
5. the experimental procedure can be completed without technical or navigation problems;
6. the overall survey length is manageable;
7. any items or instructions require revision before the main study.

No human-participant data have been collected for this project as of this version.

---

2. Pilot Design

The pilot will use the same basic structure as the planned main experiment.

Design

Five-condition between-subjects design.

Participants will be assigned to one of the following conditions:

1. Control
2. Behavior-Based
3. Review-Based
4. Product-Feature-Based
5. Mixed-Source

Each participant will view only one experimental condition.

---

3. Recommended Pilot Stages

Stage 1 — Internal Functionality Test

Before recruiting a broader pilot sample, approximately **5–10 testers** may be used to evaluate basic functionality and comprehension.

These testers are not used to evaluate the substantive hypotheses.

The internal test should evaluate:

- whether all pages appear in the correct order;
- whether condition assignment works correctly;
- whether only one condition is shown to each participant;
- whether instructions are understandable;
- whether all response options function correctly;
- whether responses are recorded correctly;
- whether the behavioral-reliance decision is stored correctly;
- whether the survey works on both desktop and mobile devices;
- whether any wording is confusing or unnatural.

Participants may be asked to provide brief qualitative feedback after completing the survey.

Stage 2 — Manipulation and Measurement Pilot

After basic functionality has been confirmed, conduct a small pilot using participants who have not previously seen the experimental materials.

Suggested Initial Target

Approximately:

N = 50–75

or roughly:

10–15 participants per experimental condition

This sample is intended primarily for detecting obvious stimulus, manipulation, measurement, or implementation problems.

It should not be treated as adequately powered for testing the main theoretical hypotheses.

---

4. Participant Eligibility

For the initial pilot, participants should ideally:

- be adults;
- have experience with online shopping;
- be able to read and understand English;
- not have previously reviewed the experimental materials.

Participants do not need to be experts in AI or recommendation systems.

Any additional eligibility criteria for the formal study should be determined before recruitment.

---

5. Pilot Procedure

Participants will proceed through the experiment in the following order:

1. Informed consent
2. Common shopping scenario
3. Random assignment to one experimental condition
4. AI recommendation stimulus
5. Perceived Relevance
6. Perceived Source Credibility
7. Situational Privacy Concern
8. Trust
9. Reliance Intention
10. Behavioral Reliance decision
11. Manipulation Checks
12. Demographic/control questions
13. Debrief

The core experimental flow should remain consistent with the planned main study.

---

6. Manipulation-Check Evaluation

The most important pilot objective is to determine whether participants correctly perceive the intended recommendation source.

Expected patterns are:

Control

Participants should generally report relatively low recognition of any specific information source.

Behavior-Based

Participants should score highest on:

MC1 — The recommendation was based on my previous browsing or shopping behavior.

Review-Based

Participants should score highest on:

MC2 — The recommendation was based on information from other consumers' reviews.

Product-Feature-Based

Participants should score highest on:

MC3 — The recommendation was based on the product's features and attributes.

Mixed-Source

Participants should recognize multiple information sources and should score particularly high on:

MC4 — The recommendation combined multiple sources of information.

Participants in the Mixed-Source condition should also tend to report relatively high recognition of behavioral, review, and product-feature information.

---

7. Manipulation Success Criteria

Because the pilot sample will be relatively small, manipulation quality should not be judged solely by statistical significance.

The following evidence should be examined together:

A. Direction of Means

Each single-source condition should show the highest or clearly elevated mean on its intended source-recognition item.

B. Separation Between Conditions

The intended source condition should be meaningfully distinguishable from unrelated conditions.

For example:

Behavior-Based participants should perceive substantially more use of behavioral information than Review-Based or Product-Feature-Based participants.

C. Participant Feedback

Participants should generally be able to describe the explanation source in a way consistent with the assigned condition.

D. Confusion Patterns

If participants repeatedly confuse two conditions, the wording of those conditions should be reconsidered.

8. Measurement Evaluation

The pilot should also evaluate the performance of the multi-item scales.

The following constructs will be examined:

- Perceived Relevance
- Perceived Source Credibility
- Situational Privacy Concern
- Trust
- Reliance Intention

For each scale, inspect:

- item distributions;
- missing responses;
- item-total relationships;
- whether any item appears confusing or inconsistent;
- preliminary internal consistency.

Cronbach's alpha and/or McDonald's omega may be calculated as diagnostic indicators.

A value around **.70 or higher** may be treated as a useful preliminary benchmark, but reliability estimates from a small pilot should not be interpreted mechanically.

---

9. Construct Separation

Special attention should be given to potentially overlapping constructs.

In particular:

Perceived Relevance vs. Trust

Participants should not interpret relevance simply as another way of saying that the AI is trustworthy.

Source Credibility vs. Trust

Source credibility refers to the perceived quality and reliability of the information supporting the recommendation.

Trust refers to confidence in the AI shopping assistant and its recommendation.

If these constructs produce nearly identical response patterns or extremely high correlations, item wording and construct definitions should be reviewed before the main study.

---

10. Behavioral Reliance Check

The pilot should verify that the behavioral decision produces meaningful variation.

Participants choose between:

- accepting the AeroSound X5 recommendation; or
- continuing to search for another product.

If nearly all participants select the same option regardless of experimental condition, the decision task may be too easy or insufficiently consequential.

This would indicate that the behavioral-reliance measure should be reconsidered before the main study.

---

11. Survey Completion and Usability

Record or observe:

- completion rate;
- completion time;
- unusually fast responses;
- unanswered items;
- navigation problems;
- participant confusion;
- mobile-display problems.

As a practical design goal, the experiment should feel short enough that participants can complete it carefully without substantial fatigue.

If participants consistently report that the survey is repetitive or unnecessarily long, items should be reviewed before adding any new measures.

---

12. Qualitative Pilot Feedback

For internal testers, an optional feedback section may include:

Q1

Was anything in the study confusing or unclear?

Q2

What information did you think the AI used to make its recommendation?

Q3

Did any question seem repetitive or difficult to answer?

Q4

Did anything about the recommendation or interface seem unrealistic?

Q5

Do you have any other comments about the study?

These feedback questions should be used for pilot refinement and should not be included as primary measures in the main experiment.

---

13. Revision Triggers

The experimental materials should be revised before the main study if one or more of the following problems appear.

Manipulation Failure

Participants cannot reliably distinguish the intended explanation sources.

Condition Confusion

Two or more conditions are repeatedly interpreted as equivalent.

Unintended Privacy Manipulation

The Behavior-Based condition is perceived as unusually invasive primarily because of extreme or unnatural wording rather than because behavioral data are being used.

Mixed-Source Overload

Participants interpret the Mixed-Source condition primarily as longer or more complicated rather than as combining multiple information sources.

Scale Problems

One or more items are consistently misunderstood or behave inconsistently with the rest of the construct.

Ceiling or Floor Effects

Most participants give nearly identical extreme responses on an outcome.

Behavioral Decision Failure

Almost everyone accepts or rejects the AI recommendation regardless of condition.

Procedure Problems

Participants encounter broken navigation, unclear instructions, or inconsistent stimulus presentation.

---

14. What Should Not Be Concluded from the Pilot

The pilot should not be used to claim that:

- one explanation source definitively increases trust;
- behavior-based explanations cause privacy concern in the population;
- review-based explanations are superior to other explanations;
- mixed-source explanations are better or worse overall;
- the proposed mediation pathways have been confirmed.

Pilot results primarily inform:

manipulation quality, measurement quality, usability, and study feasibility.

The main theoretical hypotheses should be evaluated using the adequately powered main study.

---

15. Ethics and Recruitment

Any recruitment or data collection involving human participants should follow the applicable institutional research-ethics requirements.

Before launching the formal study, determine whether institutional review, exemption, or other approval is required.

The pilot should not be represented as a completed human-subjects study until the applicable requirements have been addressed.

---

16. Pilot Decision Rule

After reviewing the pilot results, the project will proceed through one of three paths.

Proceed

The manipulations are clearly understood, measures perform reasonably, and no major implementation problems are identified.

→ Freeze stimuli and measures and prepare the main study.

Revise

The overall design is viable, but specific stimuli, questions, or implementation details require modification.

→ Revise only the affected components and conduct a targeted second pilot if necessary.

Redesign

Participants do not distinguish the central explanation-source manipulation, or the task does not produce meaningful responses.

→ Reconsider the experimental manipulation before moving to the main study.

---

17. Current Project Status

Completed:

- Conceptual Model
- Five-Condition Experimental Design
- Stimuli v2
- Manipulation Checks v1
- Hypotheses v1
- Measurement v1
- Experiment Spec v1
- Analysis Plan v1
- Power Analysis v1
- Experiment Flow v1
- Pilot Protocol v1

Next milestone:

Pilot-ready implementation

The experiment can next be transferred to an appropriate survey or experimental platform, internally tested, and refined before participant recruitment.